Program - 3

Loading of a dataset into a DataFrame and displaying its structure

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Aim

Develop a Python program to load a dataset into a DataFrame and display its structure, including shape, data types, feature names, and summary statistics.

Algorithm

1. Start
2. Import the Pandas library.
3. Read the students.csv file using pd.read_csv().
4. Store the data in a DataFrame named df.
5. Display the complete student dataset.
6. Find and display the shape of the dataset.
7. Display the number of rows and number of columns.
8. Display the feature/column names using df.columns.
9. Display the data types of each column using df.dtypes.
10. Display detailed DataFrame information using df.info().
11. Generate and display summary statistics using df.describe().
12. Display the first five student records using df.head().
13. Display the last five student records using df.tail().
14. Stop.

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```
import pandas as pd
# Load CSV file into a DataFrame
df = pd.read_csv("students.csv")

# Display the complete dataset
print("Dataset:")
print(df)

print("\nShape of the Dataset:")
print(df.shape)
print("Number of Rows:", df.shape[0])
print("Number of Columns:", df.shape[1])
print("\nFeature Names:")
print(df.columns.tolist())

print("\nData Types:")
print(df.dtypes)
print("\nDataFrame Information:")

df.info()
print("\nSummary Statistics:")
print(df.describe())

print("\nFirst Five Records:")
print(df.head())
print("\nLast Five Records:")
print(df.tail())
```